

BPX EXTRACT- DNA/RNA

Extraction Kit Manual

Catalog Number KNAE1-100

Version.1.0

For Research Use Only. Not for use in diagnostic procedures.



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WARNING! Read the Safety Data Sheets (SDSs) provided with the Kit and follow the handling instructions. Wear appropriate protective eyewear, clothing, and gloves.

1.0 Product Description

The BPX Extract-DNA/RNA Extraction Kit is designed for rapid automated or manual extraction of DNA and RNA for use in PCR and RT-PCR applications from a diverse biological sample like whole blood, feces, nasal fluids, swab samples and compatible with various transport media.

The Lysis/ binding reagents used in this kit are designed to effectively disrupt Viral, bacterial, and fungal pathogens. The magnetic microspheres used in the kit have a large available binding surface and can be fully dispersed in solution, allowing thorough Nucleic Acid (NA) binding, washing, and elution. The procedure, therefore, delivers very consistent yields of high-quality DNA/ RNA with very little sample-to-sample variation (recovery may vary depending on sample type). The BPX Extract- DNA/RNA Extraction Kit does not require Proteinase K for NA extraction and the entire procedure can be done in less than 1 hour.

2.0 Content and Storage

The BPX Extract- Total Nucleic Acid extraction Kit contains reagents to Extract NA from 100 samples.

Contents	Part Number	Amount	Storage
Lysis/Binding Solution	KNAE1-100	26.5 mL	15–30°C
Magnetic Beads	KNAE1-100-MB	1 mL	15–30°C
Wash Solution I	KNAE1-100-W1	50 mL	15–30°C
Elution Buffer	KNAE1-100-W1	10 mL	15–30°C
Cell Lysis Enzyme Mix	REMV1-2.5ML	2.5 mL	4°C

3.0 Required Materials Not Supplied

All required materials not supplied with the kit are available through biopathogenix.com or from other major laboratory suppliers (MLS).

Item Description	Vendor	Part Number
Sample Racks	MLS	
2.2 mL 96 Deep Well Plates	BioPathogenix	LS-30101
0.5 mL 96 Well Standard Plates	BioPathogenix	LS-30100
96 Tip-Comb	BioPathogenix	LS-30102
Microplate Sealer Storage Film	BioPathogenix	LS-30300
Ethanol, Molecular Biology Grade (absolute, 200 proof)	MLS	
Shaker	MLS	
Laboratory Mixer, Vortex or equivalent	MLS	
RNase Decontamination Solution	MLS	
Proteinase K	BioPathogenix	S-PROK-20MG-1
PK Buffer	BioPathogenix	S-PKB-7ML-1

4.0 Assay method for Nucleic Acid Extraction

4.1 Overview of the Procedure

The BPX extract- DNA/RNA Extraction Kit employs enzymatic digestion (for bacteria and fungi) followed by mechanical disruption of samples with ferrimagnetic beads in a guanidinium thiocyanate-based solution. This reagent system rapidly releases NA while simultaneously inactivating nucleases in the sample matrix. The magnetic beads with optimized silica-like chemistry ensure rapid magnetic mobility and efficient isolation of NA. The beads/NA complexes are immobilized on magnets and washed to remove proteins and other contaminants. A second wash solution is used to remove residual binding solution, then finally NA is eluted in a small volume of low-salt buffer.

4.2 General Guidelines for Nucleic Acid Extraction

- Before working with RNA, clean the lab bench and pipettors with an RNase decontamination solution.
- Wear laboratory gloves for this procedure; they protect you from the reagents, and they protect the RNA from nucleases that are present on skin.
- Use RNase-free Pipette tips to handle the kit reagents and avoid putting used tips into the reagent containers.
- Perform all steps at room temperature (20–25°C).
- If there are precipitates in any of these solutions, due to very low room temperature, or exposure at transport, warm them at 37°C and gently mix to dissolve precipitates. Avoid creating bubbles.

4.3 Before You Begin

BPX Extract – total Nucleic Acid Extraction Kit contains reagents for automated/manual extraction of NA from 100 samples for a specimen volume of 200 µL .

Prepare the Lysis/Binding Bead Mix- Lysis/Binding Bead Mix is prepared by either transferring entire content of magnetic beads to Lysis/binding reagent bottle for the extraction of total NA acid from 100 samples or prepare Lysis/ Binding Bead Mix for the number of required extractions according to the following table.

Component	Volume per sample
Lysis/binding solution	265 µL
Magnetic beads	10 µL
Total volume per sample	275 µL

IMPORTANT! Magnetic beads will usually be settled in the bottom of the tube. Gently pipette in and out several times to mix the magnetic beads and pipette out required volume of magnetic beads to Lysis binding solution. Keep this bottle on a shaker till use or mix them well before usage.

Prepare Lysis/Binding Bead Mix just enough for one day

Prepare Wash solution II and III. Wash solution II and III is 80% ethanol. Prepare 80% ethanol from 100% absolute ethanol and Nuclease free water.

5.0 Automated DNA/RNA Extraction from Swab, Urine or Blood Samples

1. Prepare lysis/binding bead mix and Wash Buffer II and III as described in section 4.3.
2. Add 25 µL of cell lysis enzyme to 2.2 mL 96 Deep-well sample plate for each sample.
3. Add 200 µL of samples to the well with cell lysis enzyme.
4. Prepare wash and elution plate according to the following table.

Wash Solution I	2.2 mL 96 Deep-well	500 µL
Wash Solution II 80% Ethanol	2.2 mL 96 Deep-well	500 µL
Wash Solution III 80% Ethanol	2.2 mL 96 Deep-well	250 µL
Elution Buffer	0.5 mL 96 well Standard Plates	100 µL

5. Load the prepared plates in the respective position in the automated NA extraction instrument and start the run.
6. When prompted by the instrument, remove the sample plate, then add 275 µL premade Lysis/ Binding bead mix to each well with samples.

Important. If the beads are settled down mix them gently till it gets uniformly dispersed in the solution. To ensure even distribution of beads to all specimens or wells, gently swirl the Binding Bead Mix frequently during pipetting. The Binding Bead Mix is slightly viscous, so pipet slowly to ensure that the correct amount is added. Avoid generating bubbles with binding and wash reagents by gentle mixing or pipetting.

7. Load the sample plate back in the automated NA extraction instrument and follow the prompts on the instrument to allow the sample processing to proceed.
8. On completion of the run, remove the elution plate, and eluents are ready for immediate downstream processing, or seal elution plate using a microplate sealer storage film and store at -20°C until further use.

5.1 Manual DNA/RNA Extraction from Swab, Urine or Blood samples

1. Prepare lysis/binding bead mix and Wash Buffer II and III as described in section 4.3.
2. Add 25 μL of cell lysis enzyme to 2.2 mL 96 Deep-well sample plate for each sample.
3. Add 200 μL of samples to well with cell lysis enzyme.
4. Seal the plate with Microplate sealer storage film then shake the sealed plate at 1,000 rpm for 30 sec
5. Incubate the sealed plate at 60°C for 30 minutes then shake the plate at 1000 rpm for 5 minutes.
6. Add 275 μL of Lysis/binding bead mix to each sample well.

Important. *If the beads are settled down, mix them gently till they get uniformly dispersed in the solution. To ensure even distribution of beads to all specimens or wells, gently swirl the Binding Bead Mix frequently during pipetting. The Binding Bead Mix is slightly viscous, so pipet slowly to ensure that the correct amount is added. Avoid generating bubbles with binding and wash reagents by gentle mixing or pipetting.*

7. Seal the plate with Microplate sealer storage film then shake the sealed plate at 1,000 rpm for 2 minutes
8. Incubate the sealed plate at 65°C for 5 minutes then shake the plate at 1000 rpm for 5 minutes.
9. Place the sealed plate on the magnetic stand for 10 minutes or until all the beads have collected.
10. Keeping the plate on the magnet, carefully remove the cover, then discard the supernatant from each well without disturbing the beads.
11. Remove the plate from the magnetic stand, then add 500 μL of Wash Solution I to each well.
12. Pipet up and down to mix the beads.
13. Reseal the plate, then shake at 1,000 rpm for 1 minute
14. Place the plate back on the magnetic stand for 2 minutes, or until all the beads have collected.
15. Keeping the plate on the magnet, carefully remove the cover, then discard the supernatant from each well without disturbing the beads.
16. Repeat step 11 to step 15 using 500 μL of Wash solution II.
17. Repeat step 11 to step 15 using 250 μL of Wash solution III.
18. If there is any Wash solution II left in the well, use a 20 μL multichannel pipette to remove it.
19. Dry the beads by shaking the plate (uncovered) at 1,000 rpm for 2 minutes
20. Once the beads are dried, add 100 μL of elution buffer to each specimen, then seal the plate with Microplate sealer storage film.
21. Shake the sealed plate at 1,000 rpm for 5 minutes.

22. Place the plate in an incubator at 65°C for 10 minutes
23. Place the sealed plate on the magnetic stand for 3 minutes or until clear to collect the beads against the magnets.
24. Keeping the plate on the magnet, carefully remove the seal, transfer the eluates to a fresh standard (not deep-well) 96 well plate for downstream processing, or seal elution plate using a microplate sealer storage film and store at -20°C until further use.

Note: If bead carryover is observed, place the Elution Plate on a magnetic stand to pellet the beads, then pipette the eluate to a new 96-well plate

6.0 Automated DNA/RNA extraction from Nail fungal samples

1. Prepare lysis/binding bead mix, Wash Buffer II and III as described in section 4.3.
2. Prepare enough Proteinase K Mix digestion for nail clipping as described in the following table and store at room temperature no more than 30 minutes before use.

Component	Volume Per Sample
Proteinase K	25 µL
PK Buffer	175 µL
Total Proteinase K Mix	200 µL

3. Collect the nail clip for Nucleic acid extraction in a Microfuge tube and label them.
4. Add 200 µL of Proteinase K Mix to each sample.
5. Close the tube with sample and incubate for 30 minutes at 60-65°C.
6. Incubate these tubes at 95 °C for 5 min and spin down.
7. Add 25 µL of cell lysis enzyme to 2.2 mL 96 Deep-well sample plate for each sample.
8. Transfer the entire content (approx. 200 µL) of the proteinase K digested samples to the deep well plate.
9. Prepare wash and elution plate according to the following table.

Wash Solution I	2.2 mL 96 Deep-well	500 µL
Wash Solution II 80% Ethanol	2.2 mL 96 Deep-well	500 µL
Wash Solution III 80% Ethanol	2.2 mL 96 Deep-well	250 µL
Elution Buffer	0.5 mL 96 well Standard Plates	100 µL

10. Load the prepared plates in the respective position in the automated NA extraction instrument and start the run.
11. When prompted by the instrument, remove the sample plate, and then add 275 µL premade Lysis/ Binding bead mix to each well with samples.

Important. If the beads are settled down, mix them gently till they get uniformly dispersed in the solution. To ensure even distribution of beads to all specimens or wells, gently swirl the Binding Bead Mix frequently during pipetting. The Binding Bead Mix is slightly viscous, so pipet slowly to ensure that the correct amount is added. Avoid generating bubbles with binding and wash reagents by gentle mixing or pipetting.

12. Load the sample plate back in the automated NA extraction instrument and follow the prompts on the instrument to allow the sample processing to proceed.
13. On completion of the run, remove the Elution plate, and eluents are ready for downstream processing, or seal elution plate using a microplate sealer storage film and store at -20°C until further use.

6.1 Manual DNA/RNA Extraction from Nail Fungal Samples

1. Prepare lysis/binding bead mix, Wash Buffer II and III as described in section 4.3.
2. Prepare enough Proteinase K Mix for the digestion of nail clipping as described in the following table and store at room temperature no more than 30 minutes before use.

Component	Volume Per Sample
Proteinase K	25 µL
PK Buffer	175 µL
Total Proteinase K Mix	200 µL

3. Collect the nail clip for nucleic acid extraction in a microfuge tube and label them.
4. Add 200 µL of Proteinase K Mix to each sample.
5. Close the tube with sample and incubate for 30 minutes at 60-65°C.
6. Incubate these tubes at 95 °C for 5 min and spin down.
7. Add 25 µL of cell lysis enzyme to 2.2 mL 96 Deep-well sample plate for each sample.
8. Transfer the entire content (approx. 200 µL) of the proteinase K digested samples to the deep well plate.
9. Seal the plate with Microplate sealer storage film then shake the sealed plate at 1,000 rpm for 30 sec
10. Incubate the sealed plate at 60°C for 30 minutes then shake the plate at 1000rpm for 5 minutes.
11. Add 275 µL of Lysis/binding bead mix to each sample well.

Important. If the beads are settled down, mix them gently till they get uniformly dispersed in the solution. To ensure even distribution of beads to all specimens or wells, gently swirl the Binding Bead Mix frequently during pipetting. The Binding Bead Mix is slightly viscous, so pipet slowly to ensure that the correct amount is added. Avoid generating bubbles with binding and wash reagents by gentle mixing or pipetting.

12. Seal the plate with Microplate sealer storage film then shake the sealed plate at 1,000 rpm for 2 minutes
13. Incubate the sealed plate at 65°C for 5 minutes then shake the plate at 1000 rpm for 5 minutes.
14. Place the sealed plate on the magnetic stand for 10 minutes or until all of the beads have collected.
15. Keeping the plate on the magnet, carefully remove the cover, then discard the supernatant from each well.
16. Remove the plate from the magnetic stand, then add 500 µL of Wash Solution I to each specimen
17. Pipet up and down to mix the beads.
18. Reseal the plate, then shake at 1,000 rpm for 1 minute
19. Place the plate back on the magnetic stand for 2 minutes, or until all the beads have collected.
20. Keeping the plate on the magnet, carefully remove the cover, then discard the supernatant from each well.
21. Repeat step 16 to step 20 using 500 µL of Wash solution II.
22. Repeat step 16 to step 20 using 250 µL of Wash solution III.
23. If there is any Wash solution II left in the well, use a 20 µL multichannel pipette to remove it.
24. Dry the beads by shaking the plate (uncovered) at 1,000 rpm for 2 minutes
25. Once the beads are dried, add 100 µL of Elution Buffer to each specimen, then seal the plate with Microplate sealer storage film.
26. Shake the sealed plate at 1,000 rpm for 5 minutes.
27. Place the plate in an incubator at 65°C for 10 minutes
28. Place the sealed plate on the magnetic stand for 3 minutes or until clear to collect the beads against the magnets.
29. Keeping the plate on the magnet, carefully remove the seal, transfer the eluates to a fresh standard (not deep-well) 96-well plate, for downstream processing, or seal elution plate using a microplate sealer storage film and store at -20°C until further use.

Note: If bead carryover is observed, place the Elution Plate on a magnetic stand to pellet the beads, then pipette the eluate to a new 96-well plate

7.0 Automated DNA/RNA Extraction from Nasal Swab for Virus Detection

1. Prepare Lysis/binding bead mix and Wash Buffer II as described in section 4.3.
2. Add 200 µL of samples to 2.2 mL deep well plate.
3. Add 275 µL of Lysis/binding bead mix to each sample well.

Important. If the beads are settled down mix them gently till it gets uniformly dispersed in the solution. To ensure even distribution of beads to all specimens or wells, gently swirl

the Binding Bead Mix frequently during pipetting. The Binding Bead Mix is slightly viscous, so pipet slowly to ensure that the correct amount is added. Avoid generating bubbles with binding and wash reagents by gentle mixing or pipetting.

4. Prepare wash and elution plate according to the following table.

Wash Solution I	2.2 mL 96 Deep-well	500 µL
Wash Solution II 80% Ethanol	2.2 mL 96 Deep-well	500 µL
Wash Solution III 80% Ethanol	2.2 mL 96 Deep-well	250 µL
Elution Buffer	0.5 mL 96 well Standard Plates	100 µL

5. Load the prepared plates in the respective position in the automated NA extraction instrument and start the run.
6. On completion of the run, remove the Elution plate, and eluents are ready for immediate downstream processing, or seal elution plate using a microplate sealer storage film and store at -20°C until further use.

7.1 Manual DNA/RNA Extraction from Nasal Swab for Viral Detection

1. Prepare Lysis/binding bead mix and Wash Buffer II as described in section 4.3.
2. Add 200 µL of samples to 2.2 mL deep well plate
3. Add 275 µL of Lysis/binding bead mix to each sample well.

Important. *If the beads are settled down, mix them gently till they get uniformly dispersed in the solution. To ensure even distribution of beads to all specimens or wells, gently swirl the Binding Bead Mix frequently during pipetting. The Binding Bead Mix is slightly viscous, so pipet slowly to ensure that the correct amount is added. Avoid generating bubbles with binding and wash reagents by gentle mixing or pipetting.*

4. Seal the plate with Microplate sealer storage film then shake the sealed plate at 1,000 rpm for 2 minutes
5. Incubate the sealed plate at 65°C for 5 minutes then shake the plate at 1000 rpm for 5 minutes.
6. Place the sealed plate on the magnetic stand for 10 minutes or until all the beads have collected.
7. Keeping the plate on the magnet, carefully remove the cover, then discard the supernatant from each well.
8. Remove the plate from the magnetic stand, then add 500 µL of Wash Solution I to each specimen.
9. Pipet up and down to mix the beads.
10. Reseal the plate, then shake at 1,000 rpm for 1 minute

11. Place the plate back on the magnetic stand for 2 minutes, or until all the beads have collected.
12. Keeping the plate on the magnet, carefully remove the cover, then discard the supernatant from each well.
13. Repeat step 8 to step 12 using 750 μ L of Wash solution II.
14. If there is any Wash solution II left in the well, use a 20- μ L multichannel pipette to remove it.
15. Dry the beads by shaking the plate (uncovered) at 1,000 rpm for 2 minutes
16. Once the beads are dried, add 50 μ L of elution buffer to each specimen, then seal the plate with Microplate sealer storage film.
17. Shake the sealed plate at 1,000 rpm for 5 minutes.
18. Place the plate in an incubator at 65°C for 10 minutes
19. Place the sealed plate on the magnetic stand for 3 minutes or until clear to collect the beads against the magnets.
20. Keep the plate on the magnet, carefully remove the seal, transfer the eluates to a fresh standard (not deep-well) 96-well plate Elution plate for downstream processing, or seal elution plate using a microplate sealer storage film and store at -20°C until further use.

Note: If bead carryover is observed, place the elution plate on a magnetic stand to pellet the beads, then pipette the eluate to a new 96-well plate

8.0 Safety

WARNING! GENERAL SAFETY. Using this product in a manner not specified in the user documentation may result in personal injury or damage to the instrument or device. Ensure that anyone using this product has received instructions in general safety practices for laboratories and the safety information provided in this document.

Before using an instrument or device, read and understand the safety information provided in the user documentation provided by the manufacturer of the instrument or device.

Before handling chemicals, read and understand all applicable Safety Data Sheets (SDSs) provided with the product and use appropriate personal protective equipment (gloves, gowns, eye protection, etc).

Chemical safety

WARNING! GENERAL CHEMICAL HANDLING. To minimize hazards, ensure laboratory personnel read and practice the general safety guidelines for chemical usage, storage, and waste provided below. Consult the relevant SDS for specific precautions and instructions:

- Read and understand the Safety Data Sheets (SDSs) provided by the chemical manufacturer before you store, handle, or work with any chemicals or hazardous materials.
- Minimize contact with chemicals. Wear appropriate personal protective equipment when handling chemicals (for example, safety glasses, gloves, or protective clothing).
- Minimize the inhalation of chemicals. Do not leave chemical containers open. Use only with adequate ventilation (for example, fume hood).
- Check regularly for chemical leaks or spills. If a leak or spill occurs, follow the manufacturer's cleanup procedures as recommended in the SDS.
- Handle chemical wastes in a fume hood.
- Ensure use of primary and secondary waste containers. (A primary waste container holds the immediate waste. A secondary container contains spills or leaks from the primary container. Both containers must be compatible with the waste material and meet federal, state, and local requirements for container storage.)
- After emptying a waste container, seal it with the cap provided.
- Characterize (by analysis if necessary) the waste generated by the applications, reagents, and substrates used in your laboratory. · Ensure that the waste is stored, transferred, transported, and disposed of according to all local, state/provincial, and/or national regulations.

Biological hazard safety

WARNING! BIOHAZARD. Biological samples such as tissues, body fluids infectious agents, and blood of humans and other animals have the potential to transmit infectious diseases. Conduct all work in properly equipped facilities with the appropriate safety equipment (for example, physical containment devices). Safety equipment can also include items for personal protection, such as gloves, coats, gowns, shoe covers, boots, respirators, face shields, safety glasses, or goggles. Individuals should be trained according to applicable regulatory and company/ institution requirements before working with potentially biohazardous materials. Follow all applicable local, state/provincial, and/or national regulations. The following references provide general guidelines when handling biological samples in laboratory environment.

- U.S. Department of Health and Human Services, Biosafety in Microbiological and Biomedical Laboratories (BMBL), 5th Edition, HHS Publication No. (CDC) 21-1112, Revised December 2009; found at:
www.cdc.gov/biosafety/publications/bmbl5/BMBL.pdf
- World Health Organization, Laboratory Biosafety Manual, found at:
www.who.int/csr/resources/publications/biosafety/Biosafety7.pdf